



Ultra-fast laser scanning achieves high-strength joining of glass and Al/Al-Alloys



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Abstract

The joining of glass and metals is widely applied in many industries. Although laser welding of these materials has attracted growing interest in recent years, the considerable differences in their material properties often necessitate the use of lower laser scan speeds (<20mm/s) during the welding process to realize higher laser energy input, thereby significantly reducing processing efficiency. In this study, a chromium oxide film was deposited on the surface of the quartz glass using the laser deposition technique, followed by a millisecond laser to achieve a high-strength connection between quartz glass and aluminum at ultra-high laser scanning speeds (6000mm/s). The shear strength of the welded joints can reach 20.7MPa. The width of the joint fusion zone is only 3μm. It has been shown that the formation of (Si, Al)₂Cr during the welding process is critical for ensuring high-strength connections at ultra-high laser scanning speeds. Additionally, the prepared welded joints exhibited excellent water

resistance. The method is widely applicable and enables high-strength welding of various types of glass to aluminum and its alloys under the same laser process parameters.



Keywords

Laser welding; Laser scanning speed; Aluminum; Quartz glass

1. Introduction

Glass materials are extensively utilized in various fields, including construction, aerospace, healthcare, and electronics, owing to their excellent optical properties, corrosion resistance, heat resistance, and recyclability [[1], [2], [3]]. The application of glass materials in these fields is inevitably combined with metals. Aluminum and its alloys are lightweight, exhibit good corrosion resistance, excellent electrical conductivity, and high malleability [4,5]. The joining of glass to aluminum and its alloys is extensively employed in architecture, electronic devices, automotive, and aerospace [[6], [7], [8], [9]]. However, the significant differences in properties between the two materials in aluminum-glass joints often result in issues with conventional joining methods, such as bonding [10], anodic bonding [11], brazing [12], and ultrasonic welding [13], including high residual stresses, joint deterioration, poor sealing, and operational complexity, which limit the further applications of the two materials.

Recently, pulsed lasers have emerged as a powerful tool for joining glass and metal, characterized by a small heat-affected zone and high precision. Pulsed lasers can be categorized into short-pulse and long-pulse lasers. However, both long-pulse and short-pulse lasers typically encounter two challenges in the field of metal-glass joining. First, the welding speed is slow and the strength of the joint is low. Due to the significant differences in the properties of the two materials, a high energy input is necessary during the welding process, resulting in conventional methods typically using a low welding speed, often below 20mm/s [[14], [15], [16], [17]]. Lower welding speeds not only reduce productivity but also result in excessive heat input and decreased cooling rates, which can increase defects such as porosity and cracks in the weld, ultimately reducing weld strength.

Second, optical contact (OC) must be established before welding [18,19]. OC requires high surface quality and the application of external pressure to ensure tight surface contact [20]. Conventional processes require the sample to be surface-polished and securely clamped in a fixture before welding [[20], [21], [22]]. The use of fixtures not only increases production costs but also induces localized stress concentrations within the material and significant residual stresses, ultimately compromising weld quality.

Intermetallic compounds (IMC) exhibit high strength, exceptional toughness, and good corrosion resistance, making them crucial products in dissimilar metal welding [23,24]. The thinner IMC layer significantly enhances the strength of welded joints and reduces defects, such as porosity and cracks, during the welding process [25]. Indhu et al. [26] investigated the effect of IMC on the welding strength of dual-phase steel and aluminum alloys. The results indicate that when the thickness of the IMC is below 8 μ m, the welding strength gradually increases with thickness. Huang et al. [27] enhanced the weld strength of laser-welded 5083 aluminum alloy by using activating flux Cr₂O₃ and demonstrated that the formation of the intermetallic compound Al₈Cr₅ was the primary cause of the strength increase. However, in conventional processes, laser welding of transparent brittle materials to metals primarily produces lower strength and brittle metal oxides, thereby limiting the increase in weld strength.

Herein, we report a novel method for ultrafast high-strength laser welding utilizing laser deposition technology. We seek to form ternary intermetallic compounds during the welding process to ensure weld strength at ultra-fast laser welding speeds. A low-cost and widely used millisecond laser is employed to ablate a Cr target, depositing a chromium oxide film on the quartz glass surface, and subsequently welding the quartz glass to aluminum at a scanning speed of 6000mm/s, without the need for polishing or fixture assistance. Phase identification of the quartz glass/aluminum welded joints was performed using X-ray diffraction. The results indicate that the formation of the (Si, Al)₂Cr phase is crucial for ensuring a high-strength connection at high laser scanning speeds. The effects of varying laser scanning speeds on the microstructure and elemental composition of quartz glass/aluminum joints were investigated. The interfacial bonding mechanism of this approach is elaborated. The water resistance of the welded joints was tested. Additionally, high-strength laser welding of various types of glass to aluminum and its alloys was achieved through this method using the same laser process parameters. This research will offer new ideas and a theoretical foundation for laser welding of dissimilar materials, contributing to the expansion of its industrial application.

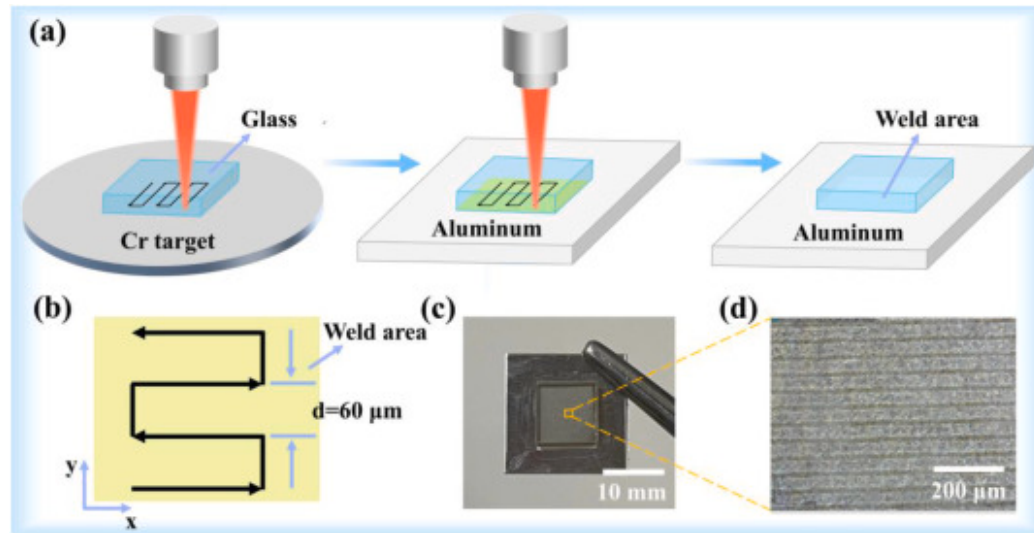
2. Experimental

2.1. Materials

The aluminum (99.7% Al) samples used in the present study had dimensions of 20mm×20mm×1 mm. The quartz glass (99.98%, SiO₂) with sizes of 10mm×10mm×3mm and 5mm×5mm×3mm were used for microstructure and mechanical property analysis, respectively. The wavelength of the millisecond laser used for the experiment was 1064nm. To confirm whether the materials achieved optical contact ($\text{gap} < \lambda/4$), the surface roughness of aluminum and quartz glass was measured using a three-dimensional laser confocal microscope (LSCM, vk-x1050, KEYENCE), as shown in [Fig. S1\(a\) and S1\(b\)](#) [15]. The arithmetic average surface roughness (Sa) of aluminum and quartz glass was measured at 1.306μm and 0.597μm, respectively, which does not meet the requirement for optical contact (0.266μm). Therefore, the sample specimens utilized in the experiments were laser welded under non-optical contact conditions. The high-purity Cr target (99.99%) was provided by Zhongnuo New Material Technology Co., Ltd.

2.2. Experimental procedure

The ms laser used in this study is a pulsed fiber laser (YLP-M30) with a maximum average output power of 30W, a pulse duration of 100ms, and a wavelength of 1064nm. The experimental procedure for laser welding utilizing laser deposition is shown in [Fig. 1\(a\)](#). First, the quartz glass was positioned on the chromium target, and the chromium target was ablated in air at a scanning speed of 6000mm/s and a laser power of 30W, depositing a chromium oxide film on the surface of the quartz glass. Second, laser transmission welding of quartz glass to aluminum was performed using the different laser process parameters. The welded area scanning pattern consisted of horizontal lines spaced 60μm apart, forming a rectangle, as shown in [Fig. 1\(b\)](#).



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Fig. 1. (a) Schematic diagram of laser welding process; (b) Scanning pattern; (c) The weld sample; (d) Micrograph of the weld zone marked with an orange rectangle in (c). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

The shear strength of the joints was tested using a tensile testing machine (990A; Company Zhiqu; China) by applying pressure at a constant displacement rate of 0.2 mm/min. The schematic diagrams of the specimen and the fixture for shear strength tests are shown in Fig. S2. The movable clamp was used to fasten the samples and ensure no deviation during the testing process. The formula for calculating the shear strength of the joint is shown in Eq. (1):

$$\sigma = \frac{F}{S} \quad (1)$$

Where σ - shear strength of the joint (MPa); F - fracture load measured in the shear test (N); S - Preset laser scanning area (mm²).

At least five samples were tested for each set of parameters to determine the average intensity values under each experimental condition.

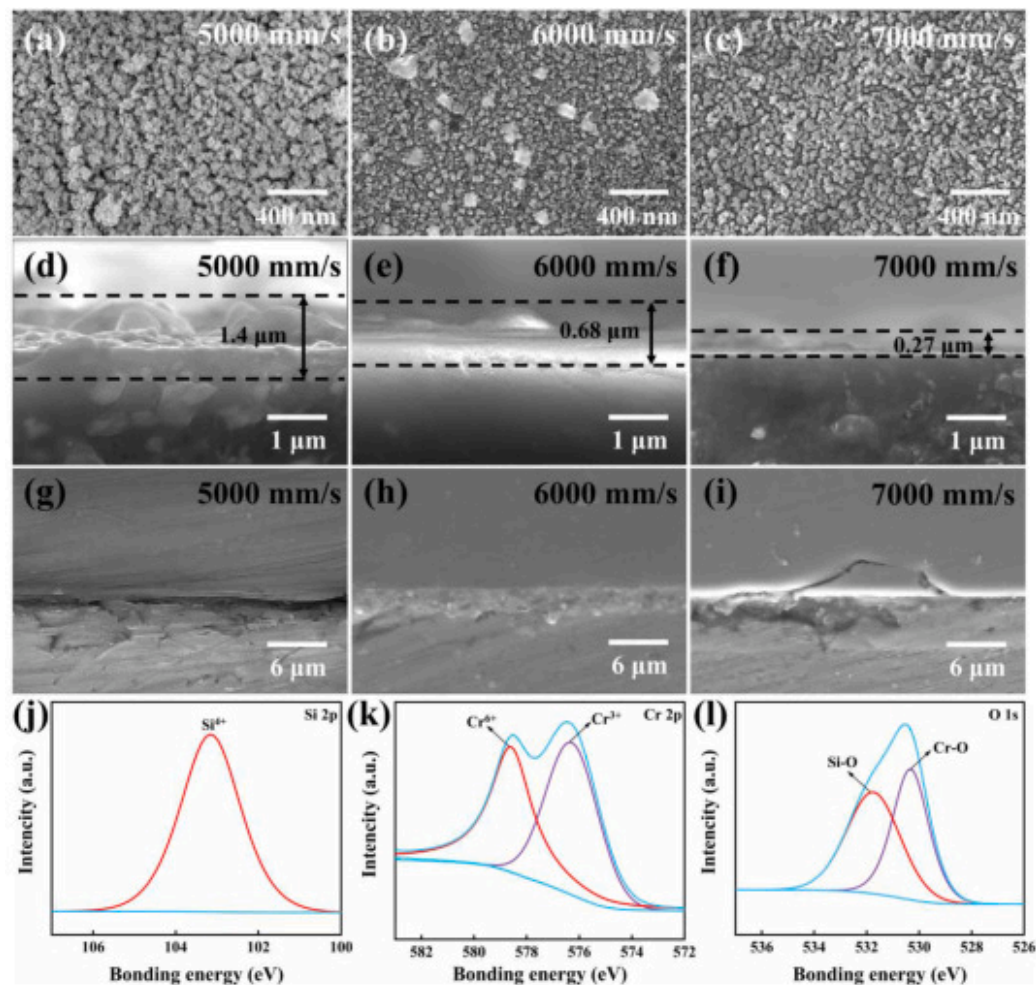
2.3. Characterisation

The macroscopic morphology of the quartz glass/aluminum interface was analyzed by Dino-lite AM4113T digital microscope. The microstructure and elemental distribution of the joint were analyzed by a scanning electron microscope (SEM, Sigma 300 CARL ZEISS) equipped with energy dispersive spectroscopy (EDS). The phase composition of the joint was analyzed by X-ray diffractometer (D8 Advance).

3. Results and discussion

The typical macroscopic morphology of millisecond laser-welded quartz glass/aluminum joints, with a power of 30W, a scanning speed of 6000mm/s, a scanning number of one, and a scanning line space of 60μm are shown in Fig. 1(c) and (d). Fig. 1(c)–(d) show the macroscopic morphology of the quartz glass/aluminum weld. The quartz glass/aluminum interface is well bonded, and the weld exhibits a periodic arrangement, free from defects such as porosity and cracks.

Fig. 2(a)–(f) show SEM images of Cr films deposited at different laser scanning speeds, and the corresponding cross-sectional EDS mappings are shown in Fig. S3. The Cr films are composed of quasi-spherical particles, with particle size decreasing progressively as the laser scanning speed increases, as shown in Fig. 2(a)–(c). The surfaces of Cr films deposited at 5000mm/s and 7000mm/s appear more uniform, whereas the film deposited at 6000mm/s exhibits a wider distribution in particle size. As shown in Fig. 2(d)–(f) and Fig. S3, the film thickness decreases with increasing scanning speed. This is attributed to the fact that a higher scanning speed leads to a lower energy density, which generates less molten material during target ablation. The elemental distributions of Cr and O are similar, indicating that the Cr film may be mainly composed of oxides. Fig. S4 presents laser confocal microscopy images of Cr films deposited at various laser scanning speeds. The film deposited at 6000mm/s exhibits the highest surface roughness ($S_a=0.41\mu\text{m}$), which is associated with a wider particle size distribution. A relatively high surface roughness promotes the mechanical interlocking effect and enhances weld strength [28].



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Fig. 2. SEM images of Cr films deposited at different laser scanning speeds: (a)–(c) surface morphology; (d)–(f) cross-sectional morphology; (g)–(i) cross-sectional morphology of the quartz glass/aluminum joint after laser welding. (j)–(l) XPS spectra of Si 2p, Cr 2p, and O 1s for Cr films deposited at a laser scanning speed of 6000 mm/s.

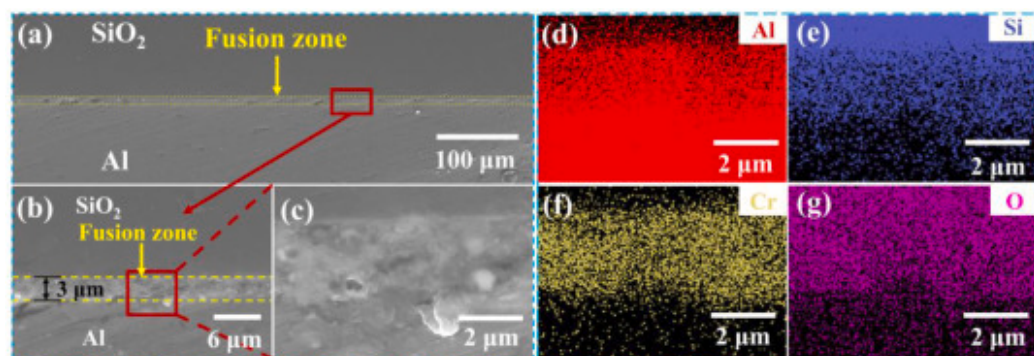
Cr films were deposited on the surface of quartz glass at different laser scanning speeds. Subsequently, the quartz glass was welded to aluminum at a scanning speed of 6000 mm/s and a laser power of 30 W. Fig. 2(g)–(i) present the cross-sectional SEM

images of the welded joints, while Fig. S5 shows their corresponding shear strengths. For Cr films deposited at 5000mm/s and 7000mm/s, large cracks and voids appeared at the quartz glass/aluminum joint after laser welding, resulting in low shear strength. In contrast, for the Cr film deposited at 6000mm/s, no voids or microcracks formed at the quartz glass/aluminum joint after laser welding, and the joint exhibited high shear strength, indicating a well-bonded interface. This is associated with the thickness and surface roughness of the Cr film [29,30].

To investigate the chemical state of the laser-deposited Cr films, those deposited at a scanning speed of 6000mm/s were analyzed using XPS (Fig. 2(j)–(l)). The characteristic Si^{4+} peak was observed in the Si 2p core-level spectrum at 103.16eV, which is attributed to SiO_2 in the substrate [31]. The main peaks in the Cr 2p XPS spectrum are observed at 576.38eV and 578.58eV, corresponding to Cr^{3+} and Cr^{6+} , respectively [32,33]. In the O 1s core energy spectrum, the peaks at 530.38eV and 531.78eV correspond to the Cr-O bond and Si-O bond, respectively [20,34]. Therefore, the Cr film is composed of Cr_2O_3 and CrO_3 . The presence of the chromium oxide layer enhances interfacial wettability, thereby improving the weld strength [35,36].

The corresponding microstructure of the quartz glass/aluminum welded joint is shown in Fig. 3(a)–(c). It can be observed that quartz glass and aluminum are well connected by this method even at ultra-fast laser scanning speeds, with no voids or microcracks formed at the interface. As shown in Fig. 3(a) and (b), a clear fusion zone is observed at the interface, exhibiting a gradient transition between SiO_2 and Al. The fusion zone exhibits a banded structure. Such a mixed morphology of quartz glass and aluminum at the interface results from a combination of factors. Firstly, recoil pressure, Marangoni convection, and capillary forces significantly influence the material flow and distribution within the melt pool. After the absorption of laser energy, the interface material undergoes thermal expansion and localized melting, resulting in recoil pressure [37]. The recoil pressure causes the molten aluminum to flow outward. As the surface temperature of the aluminum increases, energy is transferred to the glass via heat conduction, triggering the melting of the quartz glass and its mixing with the molten aluminum at the interface. The temperature gradient between the molten aluminum and glass induces the formation of Marangoni convection [38]. Marangoni convection induces the periodic migration and redistribution of molten aluminum and quartz glass. When tiny capillary channels form at the interface between molten aluminum and quartz glass, capillary forces induce the molten metal to move in different directions, further promoting the formation of the ribbon fusion zone. Additionally, the presence of a chromium oxide intermediate layer plays a crucial role in the welding process. Its excellent thermal stability and wettability not only improve the interfacial wettability between aluminum and quartz glass but also reduce the thermal expansion mismatch between the two, thereby promoting the formation of a homogeneous fusion zone [35]. Finally, the ultra-high laser scanning speed limits heat diffusion, resulting in localized thermal concentration, a smaller melt pool, and a faster melting rate [39]. The

heating-melting-solidification process of the melt pool is significantly accelerated [40]. As a result, materials are highly efficiently mixed and shaped within a short time, leading to the formation of a banded microstructure at the interface.



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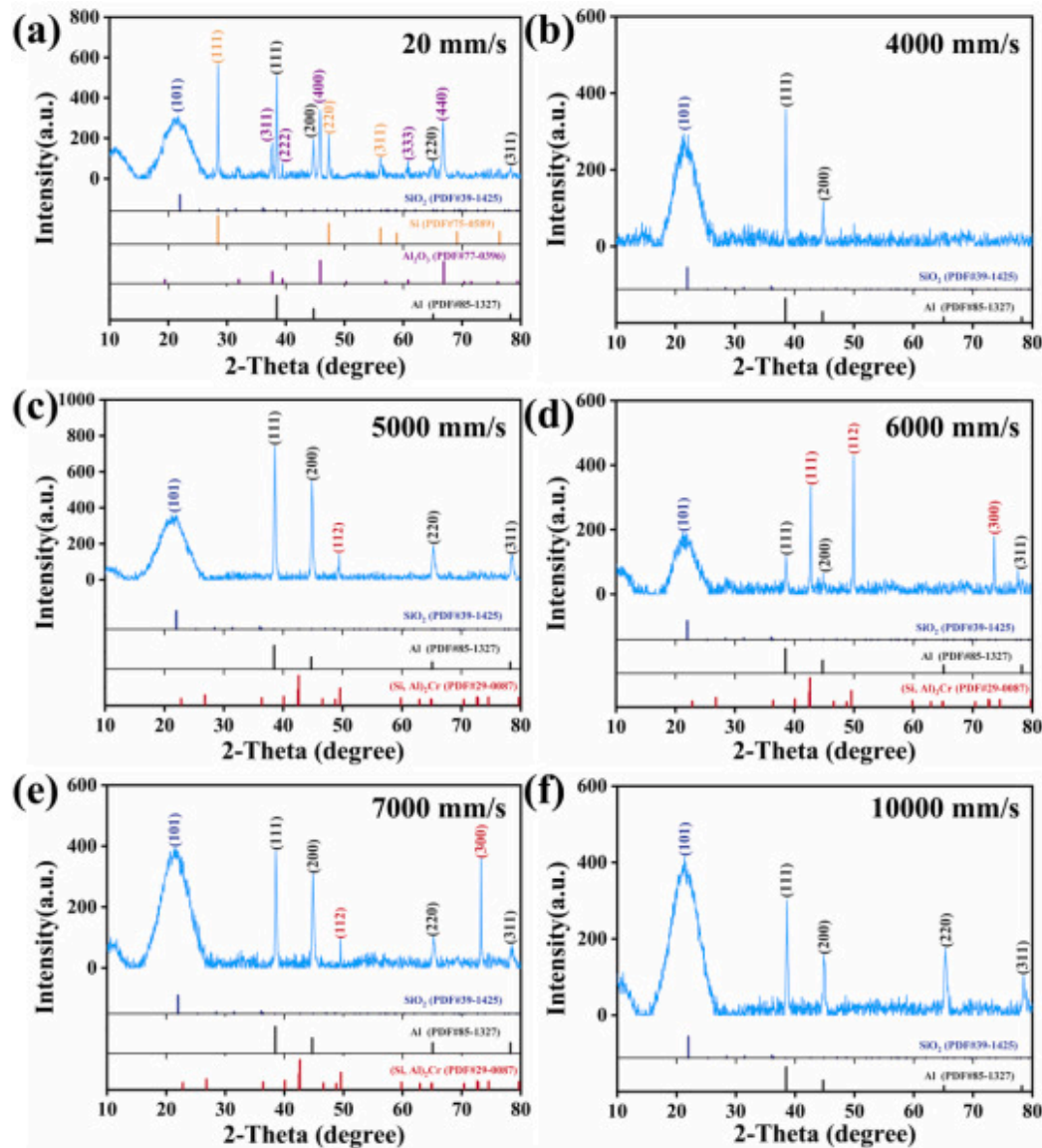
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Fig. 3. (a)–(c) Typical microstructure of the cross-section of the quartz glass/aluminum welded joint; (d)–(g) EDS surface scan of the region (c).

To identify the phase composition of the fusion zone observed in the SEM image, EDS analysis was conducted. The results are presented in Fig. 3(d)–(g) and Fig. S6. As shown in Fig. 3(d) and (e), the Al and Si elements exhibit sharp concentration gradient changes upon separate transition to the quartz glass and aluminum substrate, indicating significant diffusion and mixing of Al and Si elements within the fusion zone. The Cr element is concentrated in an area approximately 2 μm wide within the fusion zone, indicating that Cr effectively diffuses from the quartz glass into the fusion zone, as shown in Fig. 3(f). It can be observed in Fig. 3(d)–(e) and Fig. S6 that the 2-μm-wide region within the fusion zone exhibits a distinct Al, Cr, and Si-rich layer, while the remainder of the fusion zone (~1 μm wide) consists primarily of elemental Al, with the concentration of other elements decreasing dramatically, exhibiting to corresponding depleted concentrations. This result suggests that the 2-μm-wide region of the mixing zone may have formed an Al-Cr-Si ternary intermetallic compound, while the remainder is predominantly composed of molten aluminum. This phenomenon suggests that the chromium oxide layer not only plays a bridging role in promoting interfacial joining between aluminum and quartz glass but may also participate in chemical reactions during the welding process, further influencing the strength and structure of the welded joint. Furthermore, the distribution of the O element closely

resembles that of the Si element, indicating that the O element is primarily associated with molten SiO_2 , as shown in Fig. 3(e) and (g).

To further confirm the phase composition of the fusion zone and analyze the effect of different laser scanning speeds on the phase composition of the welded joints, phase identification was performed using XRD, and the results as displayed in Fig. 4. The shear strength of the quartz glass/aluminum joint at various laser scanning speeds is shown in Fig. S7.



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Fig. 4. X-ray spectra of the welded joint of the samples welded at different laser scanning speeds when the laser power is 30W, laser scanning times is 1, pulse width is 100ms, and scan pitch is $60\mu\text{m}$: (a) 20mm/s; (b) 4000mm/s; (c) 5000mm/s; (d)

6000mm/s; (e) 7000mm/s; (f) 10000mm/s.

When the laser scanning speed was 20mm/s, diffraction peaks corresponding to the cubic crystalline phases of Al (JCPDS No. 85–1327), Al_2O_3 (JCPDS No. 77–0396), and Si (JCPDS No. 75–0589) were observed as shown in Fig. 4(a). The excessively slow laser scanning speed results in excessive energy input and slow cooling rates, which disrupt the bonding between aluminum, chromium, and silicon elements and lead to stress imbalances in the weld zone. Consequently, quartz glass and aluminum cannot be effectively joined without optical contact, as shown in Fig. S7. At 4000mm/s, as shown in Fig. 4(b) and S7, only diffraction peaks corresponding to the (111) and (200) planes of Al were observed, and the shear strength of the quartz glass/aluminum joints was 3.88MPa. This is due to the still slow cooling rate, which affects the growth and alignment of crystals, resulting in low crystallinity and welding strength.

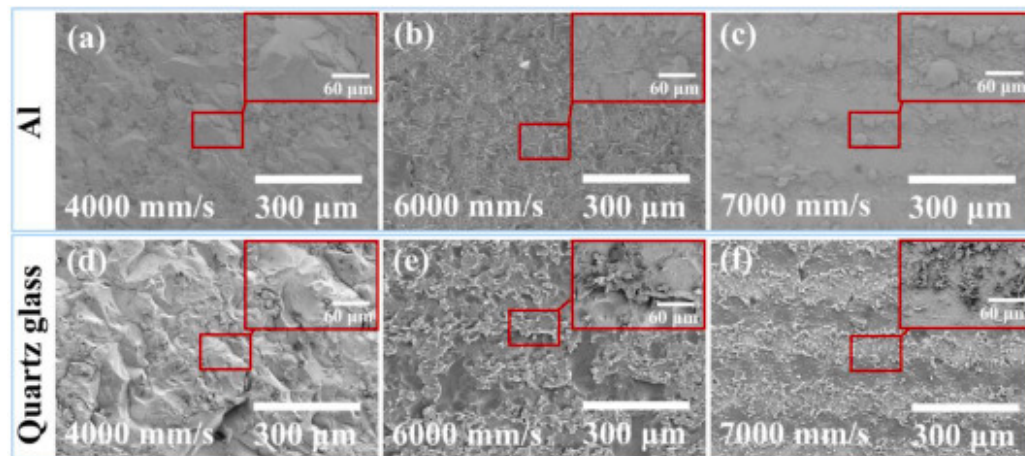
When the laser scanning speed was increased to 5000mm/s, diffraction peaks corresponding to the (111), (200), (220), and (311) planes of Al, as well as the (112) plane of $(\text{Si}, \text{Al})_2\text{Cr}$ (JCPDS No. 29–0087), were observed, as shown in Fig. 4(c). The increase in welding speed helps alleviate the imbalance between energy input and cooling rate. Consequently, the crystallinity of Al increases, and the diffraction peak of $(\text{Si}, \text{Al})_2\text{Cr}$ appears. However, due to the absence of an optimal balance between laser energy input and cooling rate, the $(\text{Si}, \text{Al})_2\text{Cr}$ phase did not fully crystallize or form, resulting in a welded joint with a shear strength of only 13.12MPa, as shown in Fig. S7. At 6000mm/s, as shown in Fig. 4(d), diffraction peaks corresponding to the (111), (112), and (300) planes of $(\text{Si}, \text{Al})_2\text{Cr}$ are observed [41]. Further increasing the laser scanning speed resulted in an optimal balance between the laser energy input and cooling rate, ensuring sufficient mixing of Al, Cr, and Si elements at the interface and an increase in the crystallinity of the $(\text{Si}, \text{Al})_2\text{Cr}$ phase. The shear strength of the quartz glass/aluminum joint reaches 20.7MPa, as shown in Fig. S7. Notably, the intensity of the diffraction peaks from aluminum decreases, indicating that the $(\text{Si}, \text{Al})_2\text{Cr}$ phase is crucial for achieving high weld strength at high laser scanning speeds. At 7000mm/s, the intensity of the diffraction peaks from the $(\text{Si}, \text{Al})_2\text{Cr}$ phase decreased, while that from Al increased. This suggests that higher scanning speeds result in insufficient energy input and excessively rapid cooling, hindering the interfacial mixing of quartz glass and aluminum. Consequently, excess molten aluminum remains at the joint interface, and incomplete crystallization of the $(\text{Si}, \text{Al})_2\text{Cr}$ phase results in a reduction of the shear strength of the welded joint to 15.96MPa (Fig. S7).

As the laser scanning speed increases to 10000mm/s, Fig. 4(f) shows that the diffraction peaks corresponding to the $(\text{Si}, \text{Al})_2\text{Cr}$ phase disappear, leaving only the peaks of the Al phase. This suggests that nearly all of the heat input at this speed was used for melting and rapid cooling, with no significant chemical reactions occurring. This results in a highly homogeneous phase structure

in the welded joint, primarily composed of Al and lacking any additional reinforcing phases. Consequently, the shear strength of the quartz glass/aluminum joint is limited to 8.64MPa.

Therefore, it can be concluded that the $(\text{Si}, \text{Al})_2\text{Cr}$ phase with good crystallinity can only form when the laser energy input is well balanced with the cooling rate. Furthermore, the $(\text{Si}, \text{Al})_2\text{Cr}$ phase is crucial for ensuring a high-strength connection of quartz glass/aluminum joints at high laser scanning speeds.

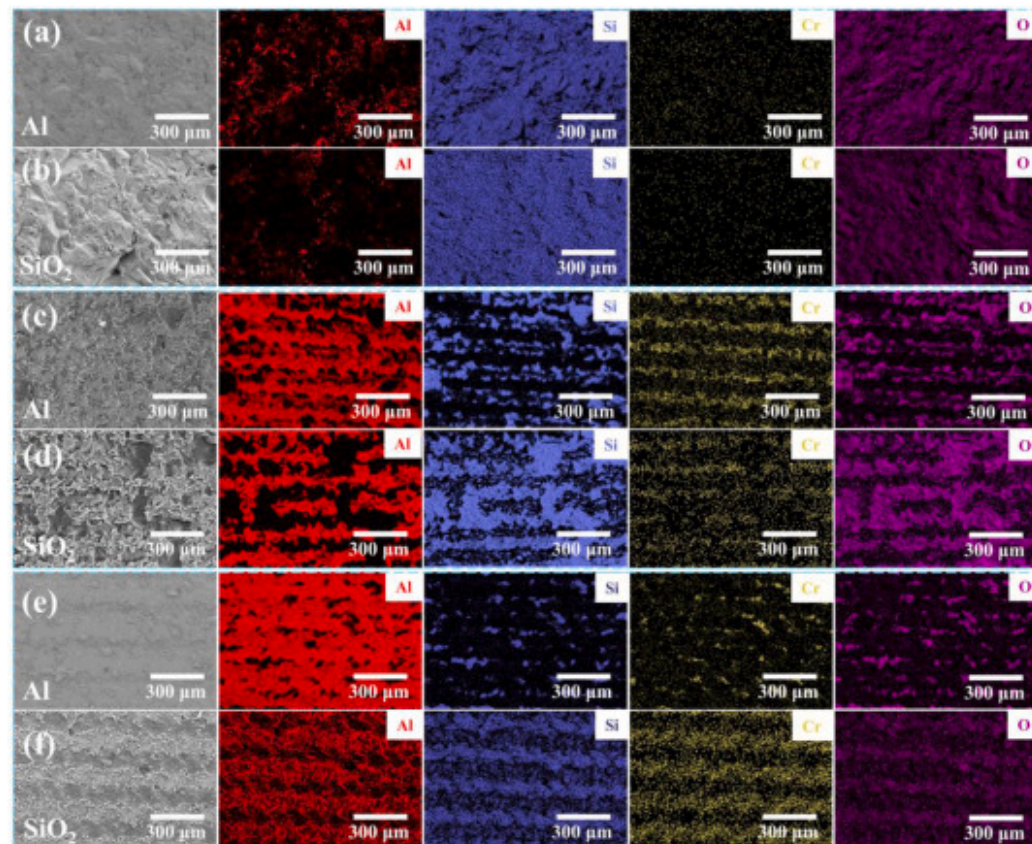
After the shear test, the fracture morphology and elemental distribution of the fracture surfaces of the quartz glass/aluminum joints were analyzed using SEM and EDS, as shown in Fig. 5, Fig. 6(a)–(f) and S8.



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Fig. 5. SEM images of aluminum/quartz glass joint fracture with different laser scanning speeds: (a)–(c) Aluminum side; (d)–(f) Quartz glass side. The red rectangles indicate areas of local magnification in the SEM images. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)



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Fig. 6. EDS surface scans of the quartz glass/aluminum joint fracture at different laser scanning speeds: (a, b) 4000mm/s; (c, d) 6000mm/s; (e, f) 7000mm/s.

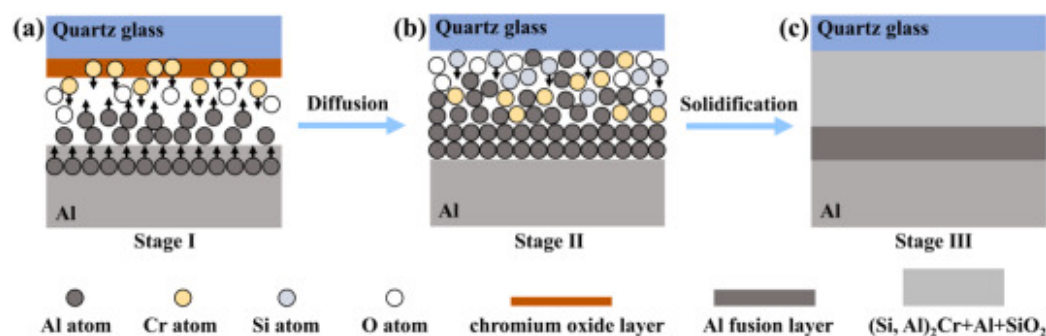
As shown in Fig. 5, Fig. 6-(b), significant cracking and spalling of the quartz glass occur on the quartz glass side, with a large amount of quartz glass adhering to the aluminum side. At the laser scanning speed of 4000mm/s, excessive laser energy tends to accumulate in the quartz glass, causing numerous cracks to form during cooling. As a result, the quartz glass easily fractures and adheres to the aluminum surface under shear forces [21]. In addition, the chromium element is irregularly distributed. This is because the high laser energy input causes the chromium oxide film to decompose and diffuse rapidly, without participating in the interfacial reaction. As shown in Fig. 6(c)–(d), when the scanning speed is 6000mm/s, the Cr element is predominantly

distributed in the weld line area on both the aluminum and quartz glass sides. This suggests that the Cr element is diffused and concentrated into the laser-irradiated area to participate in the reaction. The quartz glass/aluminum joint exhibits a typical alternating fracture morphology, while the molten material in the weld zone displays a corrugated microstructure, as shown in Fig. 5(b and e). This results from the sufficient mixing of quartz glass and aluminum at the interface [20,21]. Additionally, when the laser scanning speed was increased to 7000mm/s, the Cr elements on the aluminum side exhibited an irregular distribution, as shown in Fig. 6(e and f). This is attributed to the low energy input and rapid cooling rate resulting from the excessively fast laser scanning speed, which hindered the diffusion and mixing of the elements. The molten material on the aluminum side displays a corrugated microstructure, whereas the molten material on the quartz glass side forms an irregular network structure with numerous holes, as shown in Fig. 5(c and f). This suggests that higher laser scanning speeds caused the molten material to cool too rapidly, forming an incomplete solidification structure, which weakened the strength of the quartz glass/aluminum joint.

Fig. S8(a) and (c) reveal that the weight ratios of Al and Si are closest at the laser scanning speed of 6000mm/s, where the weight ratio of Cr reaches its maximum. This suggests that the mixing of Al and Si on the aluminum fracture surface is greatest at a laser scanning speed of 6000mm/s, while a higher amount of Cr elements diffuse into the mixing zone to participate in the reaction. As shown in Fig. S8(b) and (d), the weight ratios of Al and Si on the quartz glass fracture surface are closest at a laser scanning speed of 6000mm/s, while the weight ratio of Cr increases with increasing scanning speed. This result indicates that the degree of mixing between Al and Si on the fractured surface of the quartz glass is highest at a laser scanning speed of 6000mm/s. However, at 7000mm/s, an excess of Cr element remains on the quartz glass surface without participating in the reaction, inhibiting the formation of the $(\text{Si, Al})_2\text{Cr}$ phase, which is consistent with the XRD analysis.

The interfacial bonding mechanism can be deduced from the experimental results, interfacial phase structure, and microstructure of the quartz glass/aluminum joints, as shown in Fig. 7. The welding process is divided into three stages: the beginning of welding, interfacial reaction, and cooling solidification. In stage I, free electrons on the aluminum surface absorb the laser energy and rapidly convert it into heat, leading to a sharp rise in the surface temperature of the aluminum and localized melting [17]. Al atoms migrate upward along the temperature gradient via thermally activated diffusion [42]. Simultaneously, the chromium oxide layer absorbs a portion of the laser energy, causing Cr atoms to diffuse downward. In Stage II, as laser irradiation continues, the interfacial temperature increases further, inducing the melting of the quartz glass and significantly accelerating the elemental diffusion rate. The molten SiO_2 flows downward. The upward-migrating Al atoms rapidly react with the downward-migrating Cr and Si atoms. Additionally, due to the lower diffusion activation energy and melting point of aluminum atoms, they are more likely to migrate and diffuse at the interface, leading to the enrichment of Al atoms on the aluminum

substrate surface [43]. In stage III, laser irradiation ends with rapid cooling, and the elements and compounds in the molten state undergo solidification, forming a stable bonding interface. The reaction of Al, Si, and Cr forms the $(\text{Si}, \text{Al})_2\text{Cr}$ intermetallic compound, which, together with Al and SiO_2 , forms a composite structure upon solidification. Meanwhile, the Al-enriched region re-solidifies, forming an Al fusion layer. Thus, the joint fusion zone consists of a bilayer structure, comprising $(\text{Si}, \text{Al})_2\text{Cr} + \text{Al} + \text{SiO}_2$ and the Al fusion layer.

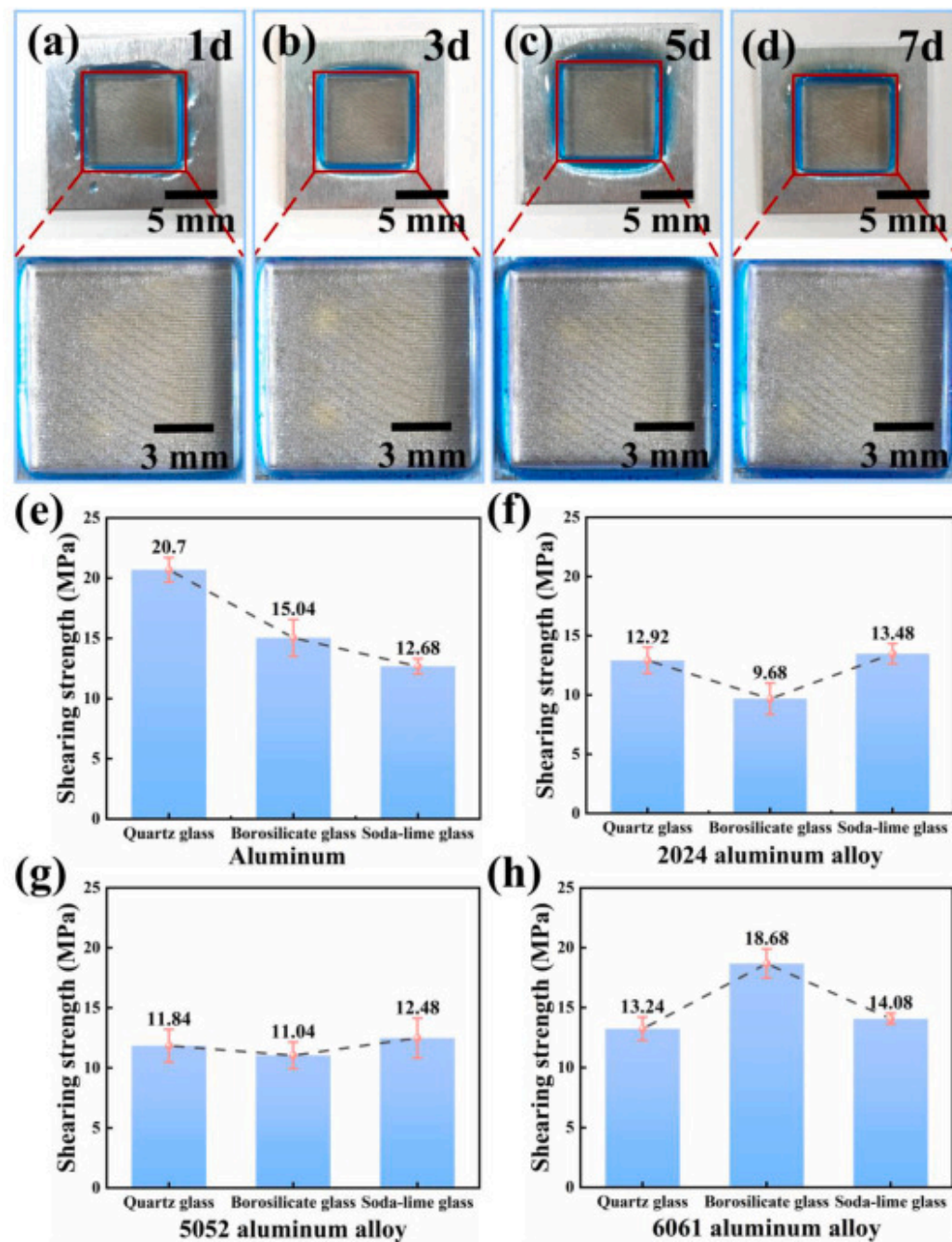


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Fig. 7. Schematic diagrams of interfacial bonding mechanism: (a) Stage I; (b) Stage II; (c) Stage III.

Water resistance tests were conducted on quartz glass/aluminum joints welded with a laser scanning speed of 6000mm/s. The welded samples were immersed in glass containers containing 200ml of methylene blue solution (20mg/L) for different immersion times (1, 3, 5, and 7 days). To minimize errors, five sets of samples were prepared for each observation time. The macroscopic morphology and shear strength of the welded samples after immersion at different times are shown in Fig. 8(a)–(d) and S9, respectively. There is no obvious intrusion of the colored solution into the weld zone is observed even after prolonged immersion, and no cracking or peeling is evident in the boundary area as shown in Fig. 8(a)–(d). As shown in Fig. S9, the shear strength of the joints decreased slightly with increasing immersion time, possibly due to slight oxidation and corrosion in the boundary region of the joints. However, the shear strength remained high, above 20MPa, even after 7 days of immersion. Consequently, samples welded using this method exhibit long-term stability in an aqueous environment.



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Fig. 8. (a)–(d) Macroscopic morphology of quartz glass/aluminum welded samples after immersion in methyl blue for different durations. Shear strength of welded joints between quartz glass, soda-lime glass, and borosilicate glass with aluminum and various aluminum alloys at a scanning speed of 6000mm/s, a laser power of 30W, a scanning line spacing of 60μm, and a single scanning pass: (e) Aluminum; (f) 2024 aluminum alloy; (g) 5052 aluminum alloy; (h) 6061 aluminum alloy. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

To verify the universality of this method, quartz glass, soda lime glass, borosilicate glass with aluminum and various aluminum alloys were laser welded using identical process parameters (laser scanning speed is 6000mm/s, laser power is 30W, laser scanning times is 1, and scan pitch is 60μm). The chemical compositions of the glass and aluminum alloys are shown in [Table S1 and S2](#). The shear strength of the welded joint is presented in [Fig. 8\(e\)–\(h\)](#). The welding process of 2024 aluminum alloy with soda-lime glass and 5052 aluminum alloy with borosilicate glass are shown in videos S1 and S2, respectively. As shown in [Fig. 8\(e\)–\(h\)](#), this method demonstrates broad applicability, and the joints of various glasses welded with aluminum and its alloys exhibit high shear strength when processed under the same parameters. This result highlights the potential of this approach to increase productivity and significantly reduce costs in industrial applications. Notably, the shear strength of the joint varies with the material combination, which is influenced by the physicochemical properties of the different materials. Process parameters can be adjusted to enhance weld strength.

[Table 1](#) presents a comparison between the results of this study and the most recent glass-to-metal laser welding research. As shown in the table, no studies have been conducted to achieve glass-to-metal connections with such fast laser scanning speeds. Although no fixtures were required, the shear strength of the welded joint reached 20.7MPa, which is much higher than that of most conventional methods. Additionally, the fusion zone width of the joint is only 3μm, narrower than that of the fusion zone in a femtosecond laser-welded glass/metal joint, which maximizes the retention of the inherent mechanical properties and microstructure of both glass and metal, reducing material property degradation due to thermal stress.

Table 1. Comparative study of laser welding for glass-metal joints.

Materials	Laser	Fixture	Scanning speed (mm/s)	Shear strength (MPa)	Width of fusion zone (μm)	Ref.
aluminum/quartz glass	ms	No	6000	20.7	3	This work

Materials	Laser	Fixture	Scanning speed (mm/s)	Shear strength (MPa)	Width of fusion zone (μm)	Ref.
Al6082/quartz glass	fs	Yes	0.1	8.94	~80	[17]
copper/silica glass	fs	No	4	17.19	35	[44]
aluminum/quartz glass	fs	Yes	0.07	25.7	~100	[21]
Kovar alloy/Soda-lime glass	fs	No	1	2	~87	[15]
Al6061/ultra-thin glass	ms	No	3	~3.54	~345	[14]
titanium alloy/borosilicate glass	ms	No	3	~3.09	~143	[45]

4. Conclusion

In this work, a high-strength joint between quartz glass and aluminum was achieved at ultra-high laser scanning speeds (6000mm/s) using laser deposition techniques. The shear strength of the welded joints can reach 20.7MPa. The width of the joint fusion zone is only 3 μm , much narrower than the fusion zone width of glass/metal joints in femtosecond laser welding reported in previous studies. The mechanical properties of the joints at different laser scanning speeds were investigated, and the evolution of the joint microstructure was analyzed using SEM, EDS, and XRD techniques. The findings indicate that the fusion zone is a bilayer structure comprising a (Si, Al)₂Cr+Al+SiO₂ layer and an Al fusion layer, with the (Si, Al)₂Cr phase being crucial to achieving a high-strength connection in quartz glass/aluminum joints at high laser scanning speeds. Additionally, the prepared welded joints exhibit excellent water resistance and maintain stability in long-term aqueous environments. This method is widely applicable and enables high-strength welding of various types of glass to aluminum and its alloys.

CRedit authorship contribution statement

Youfu Wang: Writing – original draft, Investigation. **Longfei Mi:** Writing – review & editing, Methodology, Funding acquisition. **Xiaowen Qi:** Investigation. **Xiaolong Fang:** Investigation. **Chengling Liu:** Investigation. **Hongtao Cui:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no conflict of interest.

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Appendix A. Supplementary data

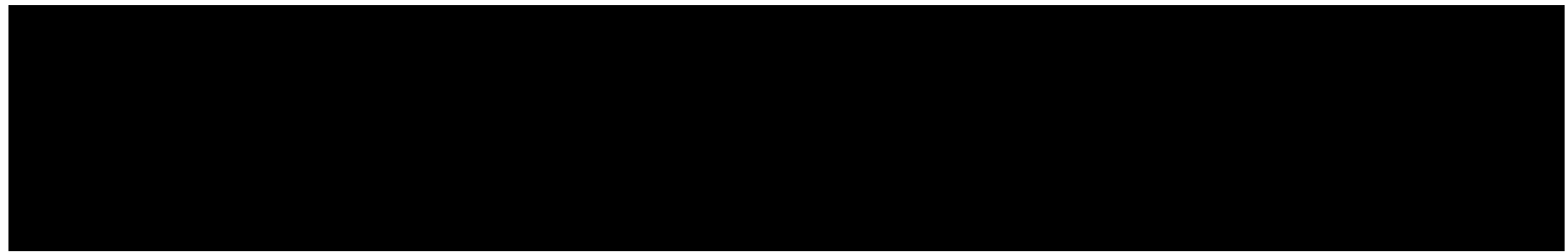
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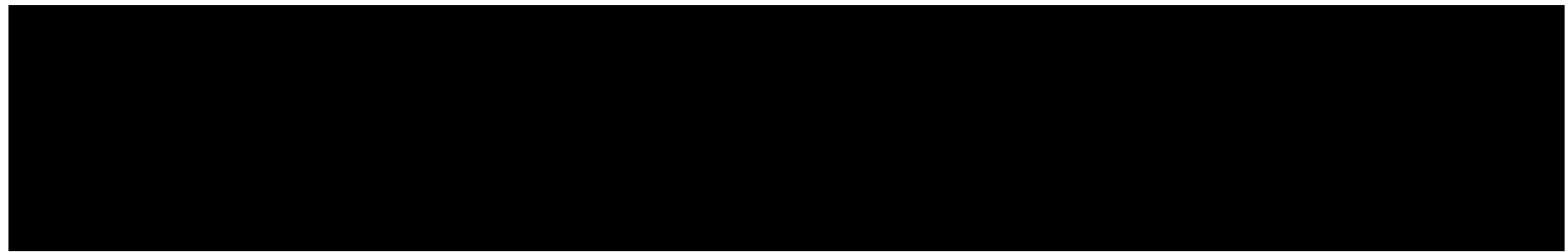
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